Reliability and Evaluation of Retrieval-Augmented Generation: A Survey

Project Proposal — Track A: Survey

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1 Background, Motivation, and Scope

Retrieval-augmented generation (RAG) conditions language generation on external evidence[3]. Retrieval can supply knowledge beyond model parameters, but irrelevant evidence, unsupported answers, and misleading citations remain reliability concerns. Reported gains may also reflect differences in generators, corpora, passage budgets, or evaluation protocols. Our survey will investigate *which reliability gains are supported by the evidence, under what conditions they hold, and which comparisons are justified*.

Gao et al.[9] organize RAG through Naive, Advanced, and Modular paradigms and the retrieval, generation, and augmentation components, alongside evaluation and benchmarks. Building on this field map, our emphasis will be a **setting-aware comparison**: separate independent design choices, connect them to evidence-use failures, and distinguish controlled comparisons from results obtained under different settings.

The core scope is **text-based, knowledge-intensive question answering and grounded answer generation**. Early retrieval and sequence-to-sequence systems provide foundations for understanding modern LLM pipelines; they will be identified as historical baselines. Adaptive text retrieval is included, while multimodal generation, graph-specific pipelines, and general autonomous-agent workflows remain outside the core comparison. Reliability covers answer correctness, support by retrieved evidence, and citation support; efficiency is a constraint on these goals.

2 Research Questions

RQ1 — Mechanisms and failures. Which retrieval and evidence-use choices improve correctness, evidence support, and citations, and under what conditions do they introduce errors?

RQ2 — Evaluation and comparability. Which metrics measure these distinct properties, and how do generator, corpus, context-budget, and judge-model differences affect comparison validity?

RQ3 — Trade-offs and applicability. When do additional passages, joint training, or adaptive retrieval justify their cost, and how far do reported gains generalize beyond tested settings?

3 Initial Taxonomy

Table 1 separates five design attributes that can coexist in one system. Evaluation is recorded in a separate evidence matrix, linking metrics and experimental settings to each reliability claim.

Table 1. Design axes and their roles in reliability analysis.

Axis	Example values	Question for analysis
Retrieval mechanism	Sparse; dense; late interaction; hybrid	Is relevant supporting evidence retrieved?
Evidence handling	Reranking; filtering; compression; source tracking	Is useful evidence retained without losing support?
Evidence fusion	Prompt concatenation; decoder fusion; latent-document marginalization	How can the generator combine and use evidence?
Parameter updates	Frozen pipeline; generator fine-tuning; joint retriever–generator training	Which component is adapted, with what supervision?
Retrieval control	One retrieval step; fixed iterative schedule; adaptive model-controlled retrieval	When is more evidence requested, and at what cost?

4 Literature and Comparison Protocol

We will search ACL Anthology, DBLP, and arXiv for RAG, grounded generation, faithfulness, citation evaluation, and adaptive retrieval (2020 to 11 October 2026). The seed list covers retrieval[2,4], retrieval-augmented modeling[1,3,5–7,10], reliability/evaluation[8,11], and an existing survey[9]; screening will add relevant 2025–2026 papers. Core papers must study external-evidence use in text generation. Foundational retriever papers will be labeled separately from complete RAG systems.

Table 2 illustrates three attributes using the original papers. The full matrix will also record retrieval and evidence handling; these entries do not imply a performance ranking.

Table 2. Initial coding of three representative systems.

System	Evidence fusion	Parameter updates	Retrieval control
RAG[3]	Marginalization over retrieved documents	Joint query-retriever and generator fine-tuning	One retrieved candidate set per input
FiD[5]	Passages encoded separately; fusion in decoder	Generator fine-tuning with an external retriever	One retrieval step before generation
Self-RAG[10]	Passage-conditioned generation and critique-based selection	Generator training with critic-produced reflection tokens	Adaptive retrieval during generation

The evidence table will record task/split, corpus, retriever, generator, training, passage/token budgets, metric, judge, and limitations. Quantitative comparisons require sufficiently aligned settings; otherwise we will report differences and compare mechanisms and applicability. Correctness, faithfulness to retrieved evidence, and citation support will be assessed separately. Cost claims will retain hardware and measurement conditions. Missing settings will be marked, and each analysis claim will link to its source result.

5 Expected Outcomes

We will deliver at least 20 relevant academic references, a multi-axis taxonomy, and an evidence-backed comparison matrix. Analysis will identify supported reliability gains, failure conditions, and evaluation gaps, informing future directions in controlled comparison, retrieval budgets, and citation support. The final report will use the official ACL LaTeX template: eight pages of body text, excluding references, with slides and a contribution statement.

6 Team Responsibilities and Timeline

Table 3. Members and owned deliverables.

Member	Responsibility and deliverable
Niu Shuai (2312659)	Coordination; retrieval, evidence-handling, and fusion coding; comparison-matrix synthesis.
Peng Kun (2313276)	Metric/benchmark matrix; reliability analysis and comparison validity.
Li Jiapu (2314007)	Screening and evidence table; update/control coding; cost and applicability analysis.
All members	Cross-review, source verification, report writing, slides, and Q&A.

Table 4. Relative schedule aligned with course milestones.

Period	Checkable milestone
Week 1	Fix scope, search strings, inclusion rules, and coding schema; cross-check pilot entries.
Weeks 2–3	Screen and annotate at least 20 papers; identify missing settings and assemble the evidence matrix.
Week 4	Consolidate the taxonomy and initial comparisons for the Survey midterm report and progress slides.
Weeks 5–6	Draft the eight-page report, supported findings, open problems, and future directions.
Week 7	Verify citations and claims, cross-review the report, and finalize slides and contributions.

Public literature and a bounded scope make the plan feasible. We will manage incomplete reporting and excessive breadth by labeling evidence gaps and retaining reliability and evaluation as the organizing questions.

References

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